CLAIMS

What is claimed is:

1. A method, comprising:

linearly passing a fiber-containing structure through at least one defect detector;

measuring at least one cross-sectional diameter of the fiber- containing structure with the defect detector to obtain at least one diameter signal of diameter versus length of the fiber-containing structure;

~~optionally~~ signal processing the diameter signal to obtain at least one signal-processed diameter signal; and

comparing ~~the diameter signal,~~ the signal-processed diameter signal~~, or combinations thereof,~~ against at least one reference signal to generate at least one surface defect signal of a surface defect output versus the length of the fiber-containing structure,

wherein:

the defect detector measures the cross-sectional diameter of the fiber-containing structure by transmitting light onto the fiber-containing structure with a projector, detecting a silhouette image of the fiber-containing structure with an optical receiver, and calculating the cross-sectional diameter based on a reduction in an amount of the light detected by the optical receiver relative to a total amount of the light transmitted by the projector; and

at least one of the defect detector is situated in series with an extrusion apparatus configured to form the fiber-containing structure, a braiding machine configured to form the fiber-containing structure, a tensioning assembly configured to apply tension to the fiber-containing structure, a finish applicator configured to apply a coating to the fiber-containing structure, a godet roll assembly configured to stretch the fiber-containing structure, a winding assembly configured to wind the fiber-containing structure onto a bobbin, or combinations thereof,

characterized in that the at least one surface defect signal comprises a slope surface defect count signal of a slope surface defect count versus the length of the fiber-containing structure, in which:

the slope surface defect count signal is generated by comparing the signal-processed diameter signal against a reference signal of a maximum first derivative of the diameter signal versus the length of the fiber-containing structure, where the signal-processed diameter signal is obtained by signal processing the diameter signal to obtain a first derivative of the diameter signal;

a slope surface defect count of zero occurs when an absolute value of the first derivative of the diameter signal is less than or equal to the maximum first derivative of the diameter signal at a particular point along the length of the fiber-containing structure; and

a slope surface defect count of greater than zero occurs when the absolute value of the first derivative of the diameter signal is greater than the maximum first derivative of the diameter signal at the particular point along the length of the fiber-containing structure, such that the slope surface defect count of greater than zero is a positive integer corresponding to a percentage of the absolute value of the first derivative of the diameter signal above the maximum first derivative of the diameter signal.

2. A method, comprising:

linearly passing a fiber-containing structure through at least one defect detector;

measuring at least one cross-sectional diameter of the fiber- containing structure with the defect detector to obtain at least one diameter signal of diameter versus length of the fiber-containing structure;

~~optionally~~ signal processing the diameter signal to obtain at least one signal-processed diameter signal; and

comparing ~~the diameter signal,~~ the signal-processed diameter signal~~, or combinations thereof,~~ against at least one reference signal to generate at least

one surface defect signal of a surface defect output versus the length of the fiber-containing structure,

wherein:

the defect detector measures the cross-sectional diameter of the fiber-containing structure by transmitting light onto the fiber-containing structure with a projector, detecting a silhouette image of the fiber-containing structure with an optical receiver, and calculating the cross-sectional diameter based on a reduction in an amount of the light detected by the optical receiver relative to a total amount of the light transmitted by the projector; and

at least one of the defect detector is situated in series with an extrusion apparatus configured to form the fiber-containing structure, a braiding machine configured to form the fiber-containing structure, a tensioning assembly configured to apply tension to the fiber-containing structure, a finish applicator configured to apply a coating to the fiber-containing structure, a godet roll assembly configured to stretch the fiber-containing structure, a winding assembly configured to wind the fiber-containing structure onto a bobbin, or combinations thereof,

characterized in that the at least one surface defect signal comprises a curvature surface defect count signal of a curvature surface defect count versus the length of the fiber-containing structure, in which:

the curvature surface defect count signal is generated by comparing the signal-processed diameter signal against a reference signal of a maximum second derivative of the diameter signal versus the length of the fiber-containing structure, where the signal-processed diameter signal is obtained by signal processing the diameter signal to obtain a second derivative of the diameter signal;

a curvature surface defect count of zero occurs when an absolute value of the second derivative of the diameter signal is less than or equal to the maximum second derivative of the diameter signal at a particular point along the length of the fiber-containing structure; and

a curvature surface defect count of greater than zero occurs when the absolute value of the second derivative of the diameter signal is greater than the maximum second derivative of the diameter signal at the particular point along the

length of the fiber-containing structure, such that the curvature surface defect count of greater than zero is a positive integer corresponding to a percentage of the absolute value of the second derivative of the diameter signal above the maximum second derivative of the diameter signal.

3. The method of claim 1 or 2, further comprising:

forming the fiber-containing structure by an extrusion process, wherein at least one of the defect detector is situated in series with the extrusion apparatus.

4. The method of claim 1 or 2, further comprising:

forming the fiber-containing structure by a braiding process, wherein at least one of the defect detector is situated in series with the braiding machine.

5. The method of claim 4, further comprising:

forming the fiber-containing structure by a braiding process,

wherein:

a defect detector array is situated in series with the braiding machine, such that the fiber-containing structure linearly passes through the defect detector array; the defect detector array comprises at least two of the defect detector situated in series;

the at least two of the defect detector are rotated relative to one another by a detector offset angle, such that different silhouette images of the fiber-containing structure are detected by the at least two of the detect detector; and

the detector offset angle ranges from 1° to less than 360°.

6. The method of claim 5, wherein the defect detector array comprises at least three of the defect detector situated in series.

7. The method of any one of claims 1-6, wherein:

the defect detector comprises two projectors for transmitting the light onto

the fiber-containing structure and two optical receivers for detecting the silhouette image of the fiber-containing structure;

the two projectors comprise a projector A, for transmitting a light A onto a surface A of the fiber-containing structure, and a projector B, for transmitting a light B onto a surface B of the fiber-containing structure;

the two optical receivers comprise an optical receiver A, for detecting a silhouette image A of the fiber-containing structure, and an optical receiver B, for detecting a silhouette image B of the fiber-containing structure;

the projector A is optically aligned with the optical receiver A, and the projector B is optically aligned with the optical receiver B;

the projector A and the projector B are rotated relative to one another by a projector offset angle, and the optical receiver A and the optical receiver B are rotated relative to one another by an optical receiver offset angle; and

the projector offset angle and the optical receiver offset angle range from 5° to 175°.

8. The method of any one of claims 1-7, wherein the defect detector is a multi-axis defect detector configured to measure at least two cross-sectional diameters of the fiber- containing structure, such that an offset angle between the at least two cross- sectional diameters ranges from 5° to 175°.

9. The method of any one of claims 1-8, further comprising:

imaging a surface of the fiber-containing structure with at least one imaging detector,

wherein:

the imaging detector images the surface by illuminating the fiber-containing structure with an imaging light, and receiving a reflected image of the surface with an imaging receiver; and

at least one of the imaging detector is situated in series with the extrusion apparatus, the braiding machine, the tensioning assembly, the finish applicator, the godet roll assembly, the winding assembly, or the combinations thereof.

10. The method of any one of claims 1-9, further comprising:

measuring a linear rate of the fiber-containing structure using at least one speedometer.

11. The method of any one of claims 1-9, further comprising:

modifying an operation of the extrusion apparatus, the braiding machine, the tensioning assembly, the finish applicator, the godet roll assembly, the winding assembly, or combinations thereof, based on the surface defect signal.